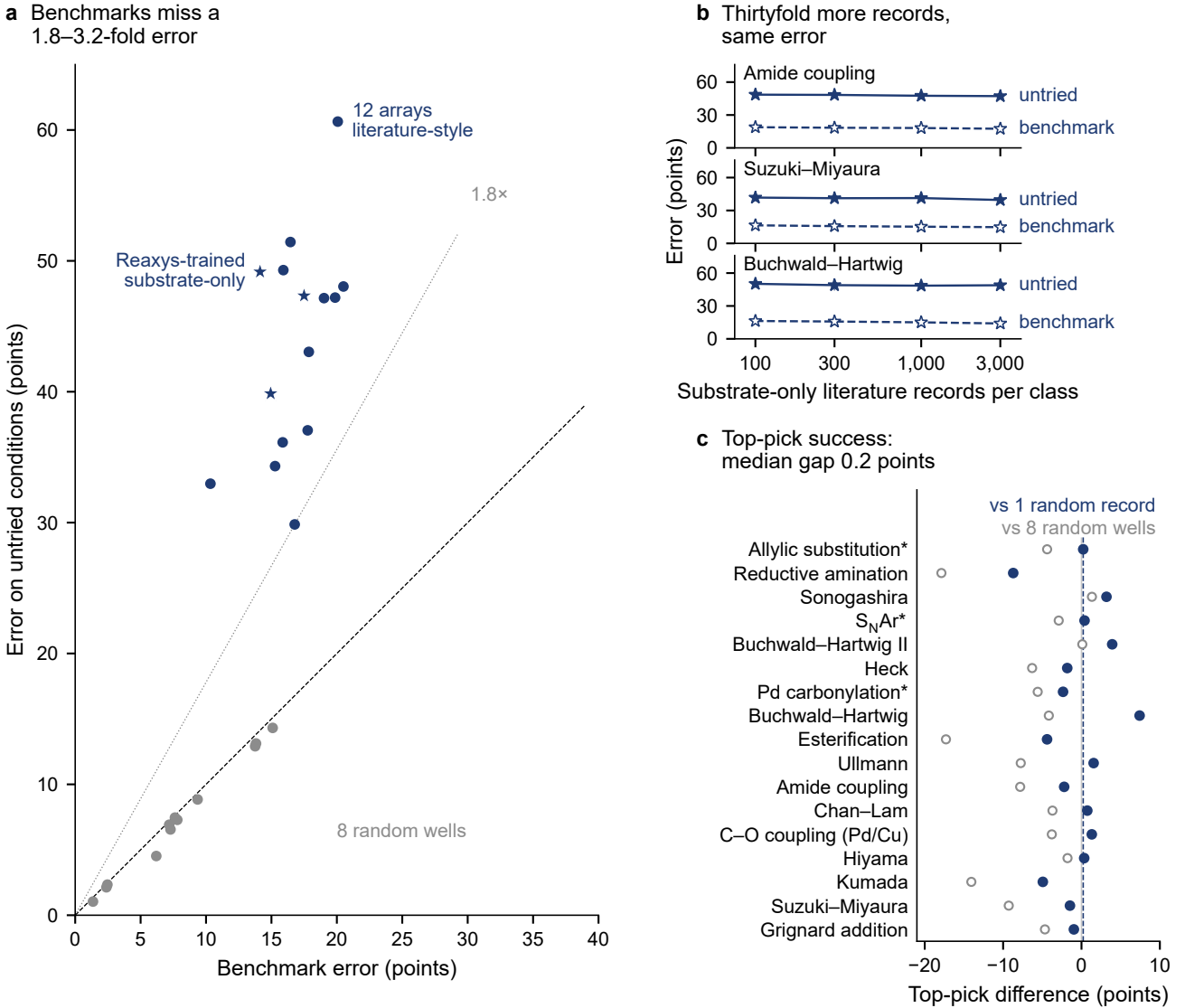
a Benchmarks miss a 1.8–3.2-fold error
12 arrays literature-style
1.8×
Reaxys-trained substrate-only
8 random wells
Error on untried conditions (points)
Benchmark error (points)
0
5
10
15
20
25
30
35
40
10
20
30
40
50
60
b Thirtyfold more records, same error
Amide coupling
Suzuki–Miyaura
Buchwald–Hartwig
untried
benchmark
Error (points)
100
300
1,000
3,000
Substrate-only literature records per class
c Top-pick success: median gap 0.2 points
vs 1 random record
vs 8 random wells
Allylic substitution*
Reductive amination
Sonogashira
SNAr*
Buchwald–Hartwig II
Heck
Pd carbonylation*
Buchwald–Hartwig
Esterification
Ullmann
Amide coupling
Chan–Lam
C–O coupling (Pd/Cu)
Hiyama
Kumada
Suzuki–Miyaura
Grignard addition
−20
−10
0
10
Top-pick difference (points)